Akash Sarode

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Education

Georgia Institute of Technology May 2028
BS in Computer Science | Threads: Intelligence, Systems & Architecture | GPA: 3.91/4.00 *Atlanta, GA*
- Core Courses: Machine Learning, Data Structures & Algorithms, Object Oriented Programming, Artificial Intelligence, Low Level Programming, Design & Analysis of Algorithms, Systems & Networks

Experience

Vanguard Sep. 2026 – Present
Software Engineering Co-op *Charlotte, NC*
- Developing backend services for Vanguard's advisory billing platform, which calculates fees for ~$400B in client assets, owning fee-profile & fee-rate APIs built in Python & TypeScript on AWS (Lambda, DynamoDB, SNS/SQS, Glue)
- Hardened data isolation between Personal & Digital Advisor platforms by building a JWT-based namespace resolver in TypeScript that scopes DynamoDB reads & writes across 3 fee-group & promotion APIs
- Identified ~170k inactive accounts with an Athena SQL query that joins fee-rate w/ account management history data

Applied Optoelectronics May 2026 – Aug. 2026
Software Engineering Intern *Duluth, GA*
- Built Python-based services to power QuantumLink: remote config & diagnostics platform monitoring millions of devices
- Developed Python-based Redis & PostgreSQL polling engine for automatic config changes, cutting device failures ~20%
- Managed virtual gateway deployments on GKE (Kubernetes) to run e2e remote download tests for ~3k virtual devices

Bits of Good: **Juno** Jan. 2026 – May 2026
Software Developer *Atlanta, GA*
- Developed a microservice API/SDK for auth/email/database/logging used by non-profit org projects in Bits of Good
- Engineered e2e file deletion feature across API gateway, S3, & Azure Blob handlers used by 5 active org project teams
- Authored a TypeScript gRPC SDK & React dashboard to streamline account approval workflows for 50+ engineers

AI Makerspace Nexus Aug. 2025 – Dec. 2025
Software Engineer *Atlanta, GA*
- Developed a RAG study tool using FastAPI & LangChain to generate quizzes from course PDFs
- Implemented RAG features including ChromaDB vectorization for semantic search across document embeddings

Urban Audio Sensing Lab Dec. 2024 – May 2026
Undergraduate Research Assistant *Atlanta, GA*
- Architected a PyTorch pedestrian classifier w/ 71.7% balanced acc using mel-spectrogram feature extraction (STFT)
- Generated pedestrian labels by running Mask2Former model on 2.9M frames & compiled detections into CSV data

Projects

Cipher Arena | **SvelteKit, Socket.IO, MongoDB, Node.js, AWS S3**
- Built a real-time multiplayer cryptogram web app for 1000+ users, featuring public & private cipher battles, detailed public profiles, a leaderboard w/ Elo rankings, & a practice mode w/ infinite drills for 10+ cipher types
- Developed a WebSocket server using Node.js & Socket.IO to manage games & ensure low-latency communication
- Implemented secure authentication using JSON Web Tokens for session management & nodemailer for email verification

SustainaView | **Gemini API, React Native, Express.js, MongoDB, Google Shopping API**
- Developed an AI mobile app using React Native, Expo, & Gemini API for eco-friendly room makeover suggestions
- Engineered a full-stack solution w/ an Express.js backend, JWT auth, MongoDB Atlas, & AWS S3 for storage
- Integrated SerpAPI & Google Shopping API for real-time price comparisons & multi-retailer product listings

Gesture Control | **JavaScript, TensorFlow.js**
- Built a Computer Vision Chrome extension for gesture based browsing, featuring customizable action mappings
- Trained a TensorFlow model on a proprietary dataset of 67k samples to classify 10 gestures w/ 93.7% accuracy

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, Java, HTML/CSS
Frameworks/Tools: Kubernetes, MongoDB, PostgreSQL, AWS, Git, Docker, LangChain, FastAPI, PyTorch
Certifications: AWS Certified Cloud Practitioner
Publication: Audio-Based Pedestrian Detection in the Presence of Vehicular Noise (arXiv:2509.19295)
Awards: FTC Robotics State Control/Software Award, 2nd at RoboTech Hackathon, REU Research Award